PAPER_958: Production Scaling v10 Benchmark (450k calc/s)

Author: Daniel T. Murphy – Star Magic / UQFF Framework **Date:** 2026-04-12 **Session:** 214 **Source:** production_scaling_v10.py (ProductionScalingV10) **Calculator:** ProductionScalingV10BenchmarkCalc (CP4 #542) **CVW:** v2.0.0 compliant

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Abstract

We benchmark the UQFF production pipeline at 450,000 calculations per second, a 12.5% improvement over the v9 baseline of 400,000. Two new kernels — BCS gap solve and spectral ladder evaluation — are added to the existing 12, bringing the total to 14 simultaneously benchmarked kernels.

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1. Scaling History

Version	Target (calc/s)	Kernels	Session
v4	100,000	4	198
v5	150,000	6	200
v6	200,000	8	204
v7	300,000	10	208
v8	350,000	11	210
v9	400,000	12	213
v10	**450,000**	**14**	**214**

2. New Kernels (v10)

kernel_bcs_gap_solve

BCS gap fixed-point iteration at $T = 4.2$ K:

$$\Delta_{n+1} = \frac{\hbar\omega_{\text{SCm}}}{2}\tanh\left(\frac{\Delta_n}{2k_B T}\right)\cdot S_{26}\cdot\frac{F_{UBi}}{F_U}$$

kernel_spectral_ladder_eval

26-state HRes spectral ladder:

$$E_n = E_0\cdot(2\pi)^{n/3}\cdot S_{26}, \quad n = 1,\ldots,26$$

3. Benchmark Architecture

All 14 kernels execute in a hot loop. Throughput is measured as:

$$\text{rate} = \frac{N_{\text{iter}}\times 14}{t_{\text{elapsed}}}$$

Target: rate $\geq 450{,}000$ calc/s.

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4. Source Data

- **File:** production_scaling_v10.py
- **Session:** 214
- **CP4 Class:** ProductionScalingV10BenchmarkCalc (#542)

References

1. Murphy, D.T. – Star Magic UQFF Framework (2024-2026)